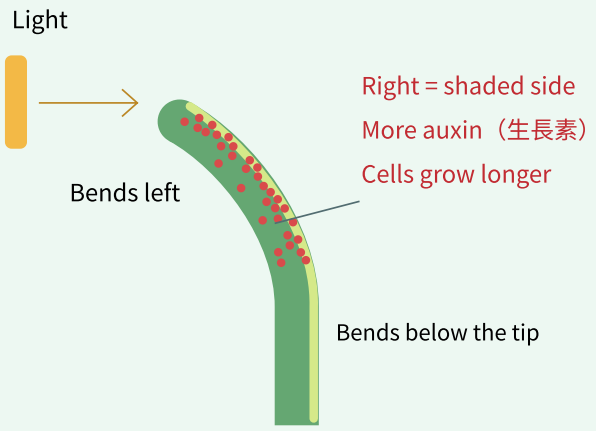
Name:_____ Class:_____ Student number:_____

From evidence to explanation

How plants grow towards light

1. Directional plant growth in response to light direction is called **"phototropism（向光性）"**; growth towards light is called "positive phototropism（向光性）".
2. In this oat coleoptile（胚芽鞘） model, **Tip** helps **sense light from one side**.
3. Other research helps us understand the substance called "substance X" in this inquiry as **Auxin（生長素）**. Auxin（生長素） is a hormone that affects plant growth; the tip can produce a growth-promoting effect that **passes downwards**.
4. With light from one side, **more auxin（生長素） is found on the shaded side, so cells on that side elongate（延長） more, making the coleoptile（胚芽鞘） bend towards light**.

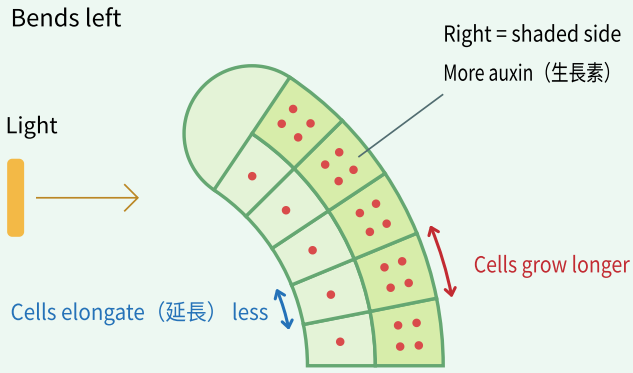
1 • Light from the left



Tip senses light → growth signal passes down

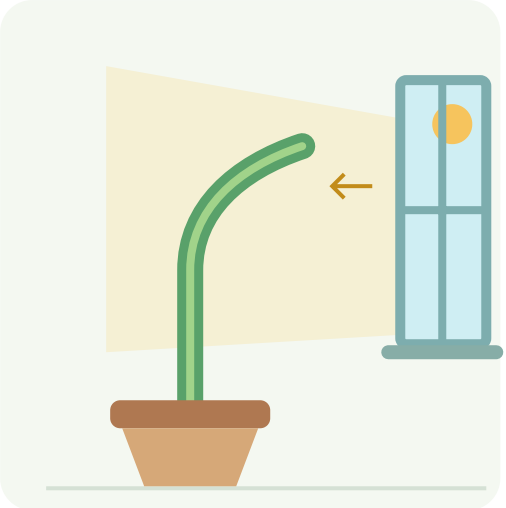
- Red dots = auxin（生長素）

2 • Light from the left: enlarged cells



More elongation（延長） on the shaded side → bends left

Learning reflection



After 24 hours

Use what you have learnt from the learning points to explain why the coleoptile（胚芽鞘） in the diagram bends to the right.